

The infinite square well potential in momentum space

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Abstract

We address an interesting problem in elementary quantum mechanics, namely how can one obtain the momentum state eigenfunctions for the infinite square well potential as a *direct* solution of the time-independent Schrödinger equation in momentum space. The conventional method for obtaining the momentum state eigenfunctions for the infinite square well potential is to take the Fourier transform of the coordinate space eigenfunctions. Unfortunately, the Schrödinger equation in momentum space for the infinite well potential is not well-defined. As such we must use a potential whose Fourier transform is well-defined and then take a limit in which the potential approaches that of the infinite well potential. We present three approaches to the problem: one based on dimensional analysis, one in which the infinite well potential is viewed as the limit of a finite well whose depth approaches infinity, and one in which the infinite well potential is viewed as the limit of a power law potential in the limit that the power approaches infinity.

I. INTRODUCTION

Quantum mechanics is a subject that continues to amaze. Problems that appear to be very simple on the surface can contain layers of complexities when one probes a bit deeper. This is evidenced in the article by F. E. Cummings entitled *The particle in the box is not simple* [1]. In considering the infinite well potential he shows that, if coordinate space eigenfunctions are used to evaluate a quantity such as $\langle \hat{k}^4 \rangle$ [$\hat{k} = \hat{p}/\hbar$ and $\hat{p} = (\hbar/i) d/dx$ is the momentum operator], the result does not agree with that obtained using momentum space eigenfunctions. The momentum space eigenfunctions are analytic functions of k in all (momentum) space, guaranteeing that the momentum space calculation of $\langle \hat{k}^4 \rangle$ is correct; on the other hand, d^4/dx^4 is *not* an analytic function of x in all coordinate space, requiring that some special care be used if $\langle \hat{k}^4 \rangle$ is to be calculated using coordinate space eigenfunctions. In coordinate space it is necessary to consider the infinite well potential as the limit of a potential function $V(x)$ for which $\langle d^4/dx^4 \rangle$ is an analytic function in all space. For example, you can model the infinite well as the limit of a steep potential having rounded edges [1] or as the limit of a finite well potential as the potential height V_0 approaches infinity. If you *use the correct coordinate space wave functions for these problems* and then take the limit that $V_0 \rightarrow \infty$, the result agrees with the momentum space calculation. In the coordinate space calculation, there are contributions to $\langle \hat{k}^4 \rangle$ that arise from what becomes the classically forbidden regions for the infinite square well potential that still contribute to the value of $\langle \hat{k}^4 \rangle$ in the limit that $V_0 \rightarrow \infty$. Of course, as $V_0 \rightarrow \infty$, the wave function vanishes in the classically forbidden region. In other words, to get the correct result for $\langle \hat{k}^4 \rangle$, you must first solve the problem for finite V_0 and *then* take the limit $V_0 \rightarrow \infty$ - the result depends on the order in which the limits are taken. An extensive discussion of the infinite well potential, including many references, is given in Ref. [2].

What is the lesson to be learned here? If you want to evaluate $\langle \hat{k}^\mu \rangle$, with $\mu \geq 4$ for the infinite well potential (or $\mu \geq 6$ for piecewise constant potentials [1]), that is, when there are discontinuities in the derivative of the potential function, momentum state eigenfunctions must be used. You might ask, why not obtain the momentum state eigenfunctions directly by solving the time-independent Schrödinger's equation in momentum (k) space for a particle

having mass m moving in a potential $V(x)$,

$$(k^2 - k_n^2) \phi_n(k) = -\frac{1}{\sqrt{2\pi}} \frac{2m}{\hbar^2} \int_{-\infty}^{\infty} dk' \tilde{V}(k - k') \phi_n(k'), \quad (1)$$

with

$$k_n^2 = \frac{2mE_n}{\hbar^2}, \quad (2)$$

$$\tilde{V}(q) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx e^{-iqx} V(x), \quad (3)$$

rather than solve Schrödinger's equation in coordinate space and then Fourier transform the eigenfunctions.

The reason that the momentum state eigenfunctions are not calculated directly is that it is very difficult to solve Eq. (1) for most *any* one-dimensional bound state potential $V(x)$. As far as we know, there are only two bound state potentials, the simple harmonic oscillator potential and the ramp potential, which can be solved easily in momentum space. There are papers describing solutions of Schrödinger's equation in momentum space for various potentials (see, for example, [3–5]), but the authors in Refs. [6] and [7] revert to taking the Fourier transform of the coordinate space eigenfunctions to obtain the momentum space eigenfunctions of the infinite square well potential. The author in Ref. [8] considers the infinite potential well as the limit of a ramp potential located within an infinite well as the ramp height goes to zero and plots the absolute square of momentum space eigenfunctions, having obtained these eigenfunctions from the Fourier transform of the coordinate space eigenfunctions.

In this paper we want to expand the known solutions of Schrödinger's equation in momentum space by obtaining a direct solution of Eq. (1) for the momentum eigenfunctions $\phi_n(k)$ of the infinite square well potential,

$$V(x) = \begin{cases} 0 & x < |a| \\ \infty & x > |a| \end{cases}. \quad (4)$$

However, we run into an immediate problem since the Fourier transform of $V(x)$ in Eq. (1) is not well-defined. The only way out of this dilemma is to start with a potential for which the Fourier transform exists and then take a limit in which the potential goes over into the infinite well potential. This is a guaranteed method for obtaining a correct result, provided the chosen potential approaches the infinite well potential in a continuous fashion as one of

the parameters in the potential is varied. Clearly you should choose the simplest potential that satisfies this criterion.

To this end, instead of using Eq. (4) for the potential, we replace it by

$$V(x) = \lim_{V_0 \rightarrow \infty} \begin{cases} 0 & x < |a| \\ V_0 & x > |a| \end{cases}, \quad (5)$$

such that Eq. (1) is replaced by

$$\lim_{V_0 \rightarrow \infty} \left[(k^2 - k_n^2) \phi_n(k) = \frac{-k_b^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk' f(k - k') \phi_n(k') \right], \quad (6)$$

where

$$k_b^2 = \frac{2mV_0}{\hbar^2}, \quad (7)$$

and

$$\begin{aligned} f(q) &= \frac{1}{\sqrt{2\pi}} \left(\int_{-\infty}^a + \int_a^{\infty} \right) dx e^{-iqx} \\ &= \frac{1}{\sqrt{2\pi}} \left[2\pi \delta(q) - 2 \frac{\sin(qa)}{q} \right]. \end{aligned} \quad (8)$$

Of course, the potential given in Eq. (5) is not the only one satisfying the necessary criterion. In Appendix A we replace that potential by $V(x) = V_0 \left(\frac{x}{a}\right)^{2r}$, where r is a positive integer, and solve Eq. (1) in the limit that $r \rightarrow \infty$. A proper solution of Eq. (6) must produce momentum state eigenfunctions and eigenenergies that are independent of V_0 in the limit that $V_0 \rightarrow \infty$.

We limit the discussion to eigenfunctions that have even parity in coordinate space; odd parity eigenfunctions can be treated in a similar manner. We might point out, however, that the odd parity momentum state eigenfunctions for the infinite well potential necessarily have the property that $\phi_n(k=0) = 0$, where n labels the eigenfunction. That is, the probability of measuring $k=0$ for a particle in such a state vanishes identically, even though the energy eigenvalues are all positive. This feature is presented as somewhat of a paradox in Ref. [9]. Of course, since the momentum and energy operators do not commute, it is possible that $\phi_n(k=0) = 0$ when $E_n \neq 0$. Before embarking on a formal solution of Eq. (1), we would like to present a solution based on dimensional analysis.

II. DIMENSIONAL ANALYSIS SOLUTION

Let us start with the free particle case, $\tilde{V} = 0$, since it will help us in obtaining the solution we are seeking. In this simple limit, equation (1) reduces to

$$(k^2 - k_n^2) \phi_n^{\text{free}}(k) = 0. \quad (9)$$

The even parity solution of Eq. (9) is

$$\phi_n^{\text{free}}(k) = \frac{1}{\sqrt{2}} [\delta(k - k_n) + \delta(k + k_n)], \quad (10)$$

with $k_n > 0$. All positive values of k_n are allowed; the spectrum is continuous and the normalization is such that

$$\int_{-\infty}^{\infty} \phi_n^{\text{free}}(k) \phi_{n'}^{\text{free}}(k) dk = \delta(k_n - k_{n'}). \quad (11)$$

For finite V_0 , Eq. (1) can be written as

$$(k^2 - k_n^2) \phi_n(k) = k_b^2 \sqrt{a/\pi} F_n(k, a, k_b), \quad (12)$$

with $F_n(k, a, k_b)$ a dimensionless function. The factor $\sqrt{a/\pi}$ is introduced into Eq. (12) since it provides the correct prescription for going from discrete energy eigenfunctions using box normalization to continuous energy eigenfunctions having the normalization given by Eq. (11). In this section we present a derivation (or an educated guess) for $\phi_n(k)$ for the infinite potential well based on some very general properties of F_n , take the limit $k_b \rightarrow \infty$, and then demand that, in the limit $a \rightarrow \infty$, $\sqrt{a/\pi} \phi_n(k)$ reverts to the solution for $\phi_n^{\text{free}}(k)$ given by Eq. (10).

According to Buckingham's Pi theorem of dimensional analysis,[10] which basically states that the laws of physics cannot depend on the choice of units, the function F_n can depend only on the three dimensionless parameters of the problem: ka , $k_b a$ and k/k_b . Equation (12) can then be written as

$$(k^2 - k_n^2) \phi_n(k) = k_b^2 \sqrt{a/\pi} G_n(ka, k_b a, k/k_b). \quad (13)$$

We first consider the limit of large (but finite) V_0 , that is, when

$$\sqrt{2mV_0/\hbar^2} = k_b \gg 1/a. \quad (14)$$

(In the real space picture, this limit corresponds to one in which the penetration depth of the particle into the classically forbidden region is much smaller than the width of the well.) In order for Eq. (13) to give a finite result in the limit $k_b a \gg 1$, we guess a solution for G_n for large $k_b a$ of the form $G_n \simeq \tilde{G}_n(ka, k/k_b)/(k_b a)^2$. With this assumption, Eq. (13) becomes

$$[(ka)^2 - (k_n a)^2] \phi_n(k) = \sqrt{a/\pi} \tilde{G}_n(ka, k/k_b). \quad (15)$$

Equation (15) retains the dependence on the potential strength through the dimensionless factor k/k_b . If the result is to be independent of V_0 in the limit that $V_0 \rightarrow \infty$, for any finite k , we must take the limit of $k/k_b \rightarrow 0$. Defining

$$g_n(ka) = \lim_{k/k_b \rightarrow 0} \tilde{G}_n(ka, k/k_b), \quad (16)$$

we find that Eq. (15) can be written as

$$\phi_n(k) = \frac{c_n \sqrt{a/\pi} g_n(ka)}{(ka)^2 - (k_n a)^2} = \frac{c_n \sqrt{a/\pi}}{2k_n a} \left(\frac{g_n(ka)}{ka - k_n a} - \frac{g_n(ka)}{ka + k_n a} \right), \quad (17)$$

where c_n is a dimensionless normalization constant.

At this point we look for a function $g_n(ka)$ for which, in the free particle limit, $a \rightarrow \infty$,

$$\sqrt{a/\pi} \phi_n(k) \sim \phi_n^{\text{free}}(k), \quad (18)$$

with $\phi_n^{\text{free}}(k)$ given by Eq. (10). In other words we want the solution to be of the form

$$\phi_n(k) = \sqrt{\frac{\pi}{2a}} \frac{1}{\pi} \left(\frac{\sin[(k - k_n)a]}{k - k_n} + \frac{\sin[(k + k_n)a]}{k + k_n} \right), \quad (19)$$

since, in the limit that $a \rightarrow \infty$,

$$\phi_n^{\text{free}}(k) \sim \sqrt{a/\pi} \phi_n(k) = \frac{1}{\sqrt{2}} [\delta(k - k_n) + \delta(k + k_n)], \quad (20)$$

where we used the fact that the delta function can be expressed as

$$\delta(x) = \frac{1}{\pi} \lim_{a \rightarrow \infty} \frac{\sin ax}{x}. \quad (21)$$

A comparison of Eqs. (17) and (19) reveals that they are not identical for arbitrary k_n ; however if we require that

$$k_n a = \left(n + \frac{1}{2} \right) \pi, \quad (22)$$

for integer n , then Eq. (19) reduces to

$$\phi_n(k) = \frac{\cos(ka)}{\sqrt{2a\pi}} \left(\frac{1}{k - k_n} - \frac{1}{k + k_n} \right), \quad (23)$$

which agrees with Eq. (17), provided

$$g_n(ka) = \cos(ka), \quad (24)$$

independent of n , and

$$c_n = \sqrt{2}k_n a = \sqrt{2} \left(n + \frac{1}{2} \right) \pi. \quad (25)$$

Note that, in the limit $a \rightarrow \infty$ there is no restriction on the values of k_n , they form a continuum.

The conclusion of this preliminary analysis is that the normalized, even parity momentum eigenfunctions for the infinite potential well are given by

$$\phi_n(k) = \frac{\sqrt{2\pi a} \left(n + \frac{1}{2} \right) \cos(ka)}{(ka)^2 - \left(n + \frac{1}{2} \right)^2 \pi^2}. \quad (26)$$

The proposed solution given in Eq. (19) is consistent with the requirement that $\phi_n(k)$ be finite for all values of k in bound state problems (see Appendix B).

Equation (26) is precisely the momentum distribution that is obtained conventionally by solving Schrödinger's equation for the infinite well in coordinate space and then taking the continuous Fourier transform to arrive at the momentum space distribution.

III. SOLUTION OF SCHRÖDINGER'S EQUATION IN MOMENTUM SPACE

We now return to Eqs. (6) and (8) and seek a solution of the integral equation

$$\lim_{(k_b a) \rightarrow \infty} \left\{ (k^2 - k_n^2) \phi_n(k) = -k_b^2 \left[\phi_n(k) - \int_{-\infty}^{\infty} dk' s(k - k') \phi_n(k') \right] \right\}, \quad (27)$$

where

$$s(q) = \frac{1}{\pi} \frac{\sin(qa)}{q}. \quad (28)$$

We look for a solution of equation (27) in the limit $(k_b a) \rightarrow \infty$, without reference to the coordinate space solution. The structure of Eq. (27) is quite general, with different potentials simply corresponding to different 'kernels' $s(q)$. Thus it is the unique properties of $s(q)$ for the square well potential that will dictate the solution.

For the particular functional form of $s(q)$ given in Eq. (28), we note two simple properties that are crucial for our derivation. The first is a “projector-like” property of our function $s(q)$:

$$\int_{-\infty}^{\infty} dq' s(q') s(q' + q) = s(q). \quad (29)$$

The second is that $\phi(q) = \cos(qa)$ [as well as $\sin(qa)$] is an eigenfunction of $s(q)$ with eigenvalue $1/2$, namely

$$\int_{-\infty}^{\infty} dq' s(q - q') \cos(q'a) = \frac{1}{2} \cos(qa). \quad (30)$$

In light of Eq. (30), we assume an even parity trial solution to Eq. (27) of the form

$$\phi_n(k) = \phi_n^{(h)}(k) \frac{2 \cos(ka)}{(k_b a)^2}, \quad (31)$$

where the function $\phi_n^{(h)}(k)$ is chosen to make the right hand side of Eq. (27) vanish, namely

$$\phi_n^{(h)}(ka) = \int_{-\infty}^{\infty} dk' s(k - k') \phi_n^{(h)}(k'a). \quad (32)$$

When the trial solution is substituted into Eq. (27) and Eq. (30) is used, we obtain

$$\phi_n^{(h)}(k) = \frac{1 \cos(ka)}{a^2 k^2 - k_n^2}, \quad (33)$$

where a term of order $1/(k_b a)^2$ was neglected on the left-hand-side of Eq. (27).

The remaining question is: does $\phi_n^{(h)}(ka)$ satisfy Eq. (32)? To see if this is the case, we rewrite Eq. (33) in the form

$$\phi_n^{(h)}(k) = \frac{1}{k_n} \left[\frac{\cos ka}{k - k_n} - \frac{\cos ka}{k + k_n} \right]. \quad (34)$$

and use the projective property given in Eq. (29). If we now choose,

$$\cos(ka) = \sin[(k - k_n)a] = -\sin[(k + k_n)a], \quad (35)$$

it is clear that Eq. (34) satisfies Eq. (32), since we obtain a solution, since, with this choice, equation (34) becomes

$$\begin{aligned} & \int_{-\infty}^{\infty} dk' s(k - k') \phi_n^{(h)}(k') \\ &= \frac{1}{k_n} \int_{-\infty}^{\infty} dk' s(k - k') \left[\frac{\cos(k'a)}{k' - k_n} - \frac{\cos(k'a)}{k' + k_n} \right] \\ &= \frac{1}{k_n} \int_{-\infty}^{\infty} dk' s(k - k') \left[\frac{\sin[(k' - k_n)a]}{k' - k_n} + \frac{\sin[(k' + k_n)a]}{k' + k_n} \right] \\ &= \frac{\pi}{k_n} \int_{-\infty}^{\infty} dk' s(k - k') [s(k' - k_n) + s(k' + k_n)] \\ &= \frac{\pi}{k_n} [s(k - k_n) + s(k + k_n)] = \phi_n^{(h)}(ka) \end{aligned} \quad (36)$$

The trial solution works, provided Eq. (35) holds, but Eq. (35) holds only if

$$k_n a = \left(n + \frac{1}{2}\right)\pi \quad (37)$$

which gives the eigenvalues for the problem.

Thus, the eigenvalues are given by Eq. (37) and the eigenfunctions can be written as

$$\phi_n(k) = c_n \sqrt{2\pi a} \left(n + \frac{1}{2}\right) \left[\frac{\cos(ka)}{(ka)^2 - (k_n a)^2} + \frac{2 \cos(ka)}{(k_b a)^2} \right], \quad (38)$$

where c_n is a normalization constant that is equal to unity in the limit that $(k_b a) \rightarrow \infty$. Equation (38) satisfies Eq. (27) and agrees with Eq. (26) in the limit that $(k_b a) \rightarrow \infty$.

IV. FOURIER TRANSFORM OF THE COORDINATE SPACE EIGENFUNCTIONS

For the potential given by Eq. (5), the even parity eigenfunctions are

$$\psi_n(x) = N \begin{cases} B e^{\kappa_n x}; & x < -a \\ \cos(k_n x) & -a < x < a \\ B e^{-\kappa_n x} & x > a \end{cases}, \quad (39)$$

where $k_n^2 = 2mE_n/\hbar^2$, $k_b^2 = 2mV_0/\hbar^2$,

$$\kappa_n^2 = \frac{2m(V_0 - E_n)}{\hbar^2} = k_b^2 - k_n^2. \quad (40)$$

and N is a normalization factor. Using the continuity of the wave function and its derivative at $x = a$, we find that

$$B = \cos(k_n a) e^{\kappa_n a} \quad (41)$$

and that the energy is determined by the equation

$$\tan(k_n a) = \kappa_n / k_n. \quad (42)$$

The Fourier transform of the eigenfunction $\psi_n(x)$ is

$$\begin{aligned} \phi_n(k) &= \frac{N}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx e^{-ikx} \psi_n(x) = \frac{1}{\sqrt{2\pi}} \int_{-a}^a e^{-ikx} \cos(k_n x) dx \\ &+ \frac{N}{\sqrt{2\pi}} \cos(k_n a) e^{\kappa_n a} \left(\int_a^{\infty} e^{-ikx} e^{-\kappa_n x} dx + \text{c.c} \right) \\ &= \phi_n^{(1)}(k) + \phi_n^{(2)}(k), \end{aligned} \quad (43)$$

where

$$\phi_n^{(1)}(k) = -N\sqrt{\frac{2}{\pi}} \cos(k_n a) \frac{\kappa_n \cos(ka) - k \sin(ka)}{k^2 - k_n^2}; \quad (44a)$$

$$\phi_n^{(2)}(k) = N\sqrt{\frac{2}{\pi}} \cos(k_n a) \frac{\kappa_n \cos(ka) - k \sin(ka)}{k^2 + \kappa_n^2}, \quad (44b)$$

"c.c" stands for "complex conjugate," and Eq. (42) was used. The function $\phi_n^{(1)}(k)$ originates from the classically allowed region and the function $\phi_n^{(2)}(k)$ from the classically forbidden region. Note that [1]

$$\phi_n(k) = \phi_n^{(1)}(k) + \phi_n^{(2)}(k) = N\sqrt{\frac{2}{\pi}} \cos(k_n a) k_b^2 \frac{k \sin(ka) - \kappa_n \cos(ka)}{(k^2 - k_n^2)(k^2 + \kappa_n^2)}. \quad (45)$$

We want to show that this wave function satisfies Schrödinger's equation in momentum space, namely

$$(k^2 - k_n^2) \phi_n(k) = -k_b^2 \left[\phi_n(k) - \frac{1}{\pi} \int_{-\infty}^{\infty} dk' \frac{\sin[(k - k')a]}{k - k'} \phi_n(k') \right], \quad (46)$$

The integrals are tabulated and, when Eq. (42) holds, it follows that

$$\phi_n^{(1)}(k) - \frac{1}{\pi} \int_{-\infty}^{\infty} dk' \frac{\sin[(k - k')a]}{k - k'} \phi_n^{(1)}(k') = 0 \quad (47)$$

and

$$\int_{-\infty}^{\infty} dk' \frac{\sin[(k - k')a]}{k - k'} \phi_n^{(2)}(k') = 0. \quad (48)$$

Using Eqs. (44b)-(48), you can then show that $\phi_n(k) = \phi_n^{(1)}(k) + \phi_n^{(2)}(k)$ is a solution of Eq. (46).

The momentum space solution we obtained is valid for any V_0 . In the limit that $V_0 \rightarrow \infty$,

$$k_n a \sim (n + 1/2)\pi [1 - 1/(k_b a)], \quad (49)$$

$$\phi_n^{(1)}(k) \sim \sqrt{2\pi a} \left(n + \frac{1}{2} \right) \frac{(-1)^{n+1}}{k^2 a^2 - [(n + 1/2)\pi]^2} \cos(ka), \quad (50)$$

$$\cos(k_n a) \sim (-1)^{n+1} (n + 1/2)\pi / (k_b a), \quad (51)$$

and

$$\phi_n^{(2)}(k) \sim \frac{\sqrt{2\pi a} (-1)^{n+1} (n + \frac{1}{2}) k_b a \cos(ka) - k a \sin(ka)}{k_b a (k^2 a^2 + k_b^2 a^2)}. \quad (52)$$

Although the second term in Eq. (52) is much less than the first for any *finite* ka , it must be retained if Eq. (48) is to be satisfied; that is, the contribution from the first term in

Eq. (52) is exactly canceled by that from the second term when the integral in Eq. (48) is carried out. We have now proven that the solution given in Eq. (38) clearly is *not* the correct eigenfunction of the finite square well in the limit of large V_0 ; it is only the correct solution in the limit that $V_0 \rightarrow \infty$. In the limit that $V_0 \rightarrow \infty$, Eq. (52) agrees with Eqs. (26) and (38).

Finally, using Eqs. (45) and (51), we see that, for $k_b a \gg 1$,

$$\phi_n(k) \sim N \sqrt{\frac{2}{\pi}} (-1)^{n+1} \frac{(n + 1/2)\pi}{a} k_b \frac{k \sin(ka) - \sqrt{k_b^2 - k_n^2} \cos(ka)}{(k^2 - k_n^2)(k^2 - k_n^2 + k_b^2)}. \quad (53)$$

It then follows that $\langle \hat{k}^2 \rangle$ is finite for any k_b since $[\phi_n(k)]^2 \sim k^{-4}$ as $k_b \rightarrow \infty$ and that $\langle \hat{k}^4 \rangle$ is finite for *finite* k_b since $[\phi_n(k)]^2 \sim k^{-6}$ for large k and finite k_b . However if we take the limit $k_b \rightarrow \infty$, then $\langle \hat{k}^4 \rangle$ diverges since $[\phi_n(k)]^2 \sim k^{-4}$ as $k_b \rightarrow \infty$.

V. CONCLUSION

We have shown that the solution of Schrödinger's equation in momentum (k) space poses some interesting challenges, since the Fourier transform of the potential does not appear to be well-defined. We used three approaches to obtain the momentum state eigenfunctions, without reverting to taking the Fourier transform of the coordinate state eigenfunctions. The three methods yield the correct momentum state eigenfunctions, but each method employed a distinct path to arrive at the final result. In the dimensional analysis approach, we took a finite well with an arbitrarily large height and looked at the limit that the well *width* went to infinity to compare with the free particle result. In the second approach, we gave a direct solution of Schrödinger's equation for a finite well in momentum space in the limit that the well height went to infinity to find the correct eigenfunctions. Finally, by approximating the potential as a power law potential (see Appendix A), we were able to solve the ensuing differential equation in momentum space in the limit that the power went to infinity. As such we have added the infinite potential well as a potential that admits to an exact solution in momentum space. Are there other potentials that also can be solved directly in momentum space? That remains to be seen.

VI. APPENDIX A - INFINITE WELL AS A LIMIT OF A $(x/a)^{2r}$ POTENTIAL

In this Appendix we study the infinite well potential $V(x)$ as the limit of the even potential $V_0(x/a)^{2r}$,

$$V(x) = V_0 \lim_{r \rightarrow \infty} \left(\frac{x}{a} \right)^{2r}, \quad (54)$$

where r is a positive integer.

In the k -space representation, the position operator is given by

$$x_{op} = i \frac{d}{dk}, \quad (55)$$

and, for a given $q = 2r$, the stationary state Schrödinger equation is

$$\left((ka)^2 + (-1)^{q/2} (k_b a)^2 \frac{d^q}{d(ka)^q} \right) \phi_n^{(q)}(ka) = (k_n a)^2 \phi_n^{(q)}(ka), \quad (56)$$

with $(k_b a)^2 = 2mV_0 a^2 / \hbar^2$. Setting

$$\xi = ka; \quad (57a)$$

$$\xi_b = k_b a; \quad (57b)$$

$$\xi_n = k_n a, \quad (57c)$$

we can recast Eq. (56) as

$$(\xi^2 - \xi_n^2) \phi_n^{(q)}(\xi) = g_n^{(q)}(\xi), \quad (58)$$

where

$$g_n^{(q)}(\xi) = -(-1)^{q/2} \xi_b^2 \frac{d^q}{d\xi^q} \phi_n^{(q)}(\xi) \quad (59)$$

We want to find $g_n^{(q)}(\xi)$ for $q \rightarrow \infty$. Consider Eq. (59) for two consecutive values of q , differing by 2. It follows from Equation (58) that

$$(\xi^2 - \xi_n^2) \phi_n^{(q+2)}(\xi) = g_n^{(q+2)}(\xi) \quad (60)$$

$$= (-1)^{q/2} \xi_b^2 \frac{d^{q+2}}{d\xi^{q+2}} \phi_n^{(q+2)}(\xi) \quad (61)$$

$$= -\frac{d^2}{d\xi^2} \left[-(-1)^{q/2} \xi_b^2 \frac{d^q}{d\xi^q} \phi_n^{(q+2)}(\xi) \right] \quad (62)$$

For very large q we expect $\phi_n^{(q)}(\xi)$ to converge to the infinite square well limit, that is,

$$\phi_n^{(q+2)}(\xi) - \phi_n^{(q)}(\xi) \sim 0, \quad (63)$$

which implies that the square bracket in Eq. (62), in the limit $q \rightarrow \infty$, approaches $g_n^{(q)}(\xi) \sim g_n^{(\infty)}(\xi)$, and we obtain:

$$\frac{d^2}{d\xi^2} g_n^{(\infty)}(\xi) = -g_n^{(\infty)}(\xi). \quad (64)$$

Since we want $g(\xi_n)$ to have an infinite number zeroes and are considering only even parity solutions, the only acceptable solution for $g_n^{(\infty)}(\xi)$ and ξ_n satisfying this criterion is

$$\begin{aligned} g_n^{(\infty)}(\xi) &= \cos \xi; \\ \xi_n &= (n + 1/2)\pi, \\ n &= 0, 1, 2, \dots \end{aligned} \quad (65)$$

The corresponding (unnormalized) solution for $\phi_n^{(\infty)}(\xi)$ is

$$\phi_n^{(\infty)}(\xi) = \frac{\cos \xi}{\xi^2 - \xi_n^2}. \quad (66)$$

However, $\phi_n^{(\infty)}(\xi)$ is not a solution of Eq. (58) since

$$(\xi^2 - \xi_n^2) \phi_n^{(\infty)}(\xi) = \cos \xi \neq -\lim_{q \rightarrow \infty} (-1)^{q/2} \frac{d^q}{d\xi^q} \frac{\cos \xi}{\xi^2 - \xi_n^2}. \quad (67)$$

In order to show that $\phi_n^{(\infty)}(\xi)$ is indeed a solution (in the limit $q \rightarrow \infty$) we proceed as follows. First rewrite $\phi_n^{(\infty)}(\xi)$ as a Taylor expansion:

$$\phi_n^{(\infty)}(\xi) = \frac{\cos \xi}{\xi^2 - \xi_n^2} \quad (68)$$

$$= \frac{1}{2\xi_n} \left(\frac{\cos \xi}{\xi - \xi_n} - \frac{\cos \xi}{\xi + \xi_n} \right) \quad (69)$$

$$= \frac{(-1)^n}{2\xi_n} \left(\frac{\sin(\xi - \xi_n)}{\xi - \xi_n} + \frac{\sin(\xi + \xi_n)}{\xi + \xi_n} \right) \quad (70)$$

$$= \frac{(-1)^n}{2\xi_n} \sum_{m=0}^{\infty} (-1)^m \frac{u^{2m} + v^{2m}}{(2m+1)!}, \quad (71)$$

with $u = \xi - \xi_n$ and $v = \xi + \xi_n$. Next, we define a truncated function, which we call $\bar{\phi}_n^r(\xi)$, given by

$$\bar{\phi}_n^{(r)}(\xi) = \frac{(-1)^n}{2\xi_n} \sum_{m=0}^{r-1} (-1)^m \frac{u^{2m} + v^{2m}}{(2m+1)!},$$

with

$$r = \frac{q}{2}. \quad (72)$$

Similarly, we define a truncated series expansion for the cosine function as

$$(\cos \xi)_r = \sum_{m=0}^{r-1} (-1)^m \frac{\xi^{2m}}{(2m)!}.$$

These truncated functions satisfy

$$\lim_{r \rightarrow \infty} \bar{\phi}_n^{(r)}(\xi) = \frac{\cos \xi}{\xi^2 - \xi_n^2}; \quad (73a)$$

$$\lim_{r \rightarrow \infty} (\cos \xi)_r = \cos \xi, \quad (73b)$$

and, since the truncated functions are polynomials of order $2r - 2$,

$$\frac{d^{2r}}{d\xi^{2r}} \bar{\phi}_n^{(r)}(\xi) = 0, \quad (74a)$$

$$\frac{d^{2r}}{d\xi^{2r}} (\cos \xi)_r = 0. \quad (74b)$$

As a trial solution we take

$$\tilde{\phi}_n^r(\xi) = \bar{\phi}_n^{(r)}(\xi) - \cos\left(\xi/\xi_b^{1/r}\right) + (\cos \xi)_r. \quad (75)$$

The factor of $1/\xi_b^{1/r}$ has been added so that when Eq. (75) is inserted into Eq. (58), Eq. (58), the left hand side becomes,

$$\{L.h.s \text{ of (58)}\} = (\xi^2 - \xi_n^2) \left(\bar{\phi}_n^{(r)}(\xi) - \cos\left(\xi/\xi_b^{1/r}\right) + (\cos \xi)_r \right), \quad (76)$$

while the right hand side is different, and given by:

$$\{R.h.s. \text{ of (58)}\} = \cos\left(\xi/\xi_b^{1/r}\right). \quad (77)$$

In other words, the trial function is not an exact solution of the differential equation for finite r .

However, in the limit $r \rightarrow \infty$, equation (76) and (77) coincide, and the trial solution becomes indeed a solution. Note that although the approximate solution reduces to the correct solution as $r \rightarrow \infty$ for any finite r , the trial solution may or may not be closely related to the true solution of the x^{2r} potential. In summary, in this section we offered an alternative route to show that the solution of the problem is of the form (66).

When properly normalized, the eigenfunction agrees with Eqs. (26) and (38) in the limit that the potential approaches an infinite well potential.

VII. APPENDIX B - FINITENESS OF MOMENTUM STATE WAVE FUNCTION

In deriving some of the results, we assumed that the momentum state eigenfunctions for bound state problems cannot diverge for any value of the momentum. To prove this, we suppose that $\phi(k)$ diverges at $k = k_0$ but that

$$\frac{1}{\sqrt{2\pi}} \int_{k_0-\epsilon}^{k_0+\epsilon} \phi(k) dk = g(k_0, \epsilon)$$

is finite for any $\epsilon > 0$. The Fourier transform of $\phi(k)$ will have a contribution of the form

$$\frac{1}{\sqrt{2\pi}} \lim_{\epsilon \rightarrow 0} \int_{k_0-\epsilon}^{k_0+\epsilon} \phi(k) e^{ikx} dk.$$

If we now let ϵ get arbitrarily small, we find that this contribution is equal to

$$\frac{1}{\sqrt{2\pi}} \lim_{\epsilon \rightarrow 0} \int_{k_0-\epsilon}^{k_0+\epsilon} \phi(k) e^{ikx} dk \rightarrow g(k_0, \epsilon) e^{ik_0 x}. \quad (78)$$

In other words there is a constant amplitude, non-vanishing contribution to the spatial wave function in all space, which violates our assumption that the eigenfunction corresponds to a bound state. As a consequence $\phi(k)$ must be finite for any k .

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